

Kyeongseon (Ella) Kim

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RESEARCH INTERESTS

How multimodal models and agents understand *abstract and structured visual representations*, including vector graphics, interfaces, and symbols. I investigate how visual form encodes meaning, how interaction reveals information, and how cultural context shapes model responses.

EDUCATION

KAIST

Ph.D. Student, School of Computing (Advisor: Prof. Tae-Hyun Oh)

Daejeon, Korea

Sep. 2024 – Present

Chung-Ang University

M.S. in Computer Science (Advisor: Prof. Joongheon Kim, now at Korea University)

Seoul, Korea

Sep. 2017 – Aug. 2019

B.S. in Computer Science and Engineering

Mar. 2013 – Feb. 2017

PUBLICATIONS

“*” denotes equal contribution

[C8] **IGG: A Benchmark for Interactive GUI Grounding under Visibility Constraints.**

K. S. Kim*, J. Son*, and T.-H. Oh.

Conference on Empirical Methods in Natural Language Processing (EMNLP 2026, Main)

Also presented at ICML 2026 Workshop on Compositional Learning: Safety, Interpretability, and Agents

[C7] **SVHalluc: Benchmarking Speech–Vision Hallucination in Audio-Visual Large Language Models.**

C. Zhang, **K. S. Kim**, C. Liu, and T.-H. Oh.

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026)

[C6] **mEOL: Training-Free Instruction-Guided Multimodal Embedder for Vector Graphics and Image Retrieval.**

K. S. Kim, S.-E. Baek, J.-M. Lee, and T.-H. Oh.

IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2026)

Accepted as a Round 1 paper (6.4% acceptance rate); Best Student Presentation Award, IPIU 2025

Also presented at ICCV 2025 Workshop on Graphic Design Understanding and Generation (GDUG)

[C5] **Zero-shot Depth Completion via Test-time Alignment with Affine-invariant Depth Prior.**

H. Lee, **K. S. Kim**, B.-K. Kwon, and T.-H. Oh.

AAAI Conference on Artificial Intelligence (AAAI 2025)

Qualcomm Innovation Fellowship 2025 winning paper

[J1] **Personalized Online Live Video Streaming Using Softmax-Based Multinomial Classification.**

K. S. Kim, D. Kwon, J. Kim, and A. Mohaisen.

Vol. 9, No. 11, Article 2297 (Applied Sciences 2019)

[C4] **Hardness on Style Transfer Deep Learning for Rococo Painting Masterpieces.**

K. S. Kim, D. Kim, and J. Kim.

International Conference on Artificial Intelligence in Information and Communication (ICAII 2019)

[C3] **Detecting Selfish Backoff Attack in IEEE 802.15.4 CSMA/CA Using Logistic Classification.**

J. Kim and **K. S. Kim**.

International Conference on Ubiquitous and Future Networks (ICUFN 2018)

[C2] **Recipient-oriented Transaction for Preventing Double Spending Attacks in Private Blockchain.**

H. Lee, M. Shin, **K. S. Kim**, Y. Kang, and J. Kim.

IEEE International Conference on Sensing, Communication, and Networking (IEEE SECON 2018)

[C1] **Self-Adaptive Machine Learning Operating Systems for Security Applications.**

K. S. Kim, D. Kwon, Y. Kim, J. Kim, and J. Kim.

IEEE International Conference on Acoustics, Speech and Signal Processing (IEEE ICASSP 2018)

[W1] **CuPS: Measuring Cultural Preference Signatures in LLM/VLM Agents and Their Steering by Profile Memories.**

K. S. Kim, G. Kim, J. Chang, H. Kim, and T.-H. Oh.

Workshop on Culture × AI: Evaluating AI as a Cultural Technology (ICML 2026 Workshop)

COMMUNITY LEADERSHIP & EVENT ORGANIZATION

AGI Korea (formerly TensorFlow Korea) – Founder & Organizer (66K+ members)	2020 – Present
Google TensorFlow User Group (TFUG) – Organizer (Global)	2020 – Present
TensorFlow Everywhere – General Organizer (Global)	Feb. 2021
Upstage × Naver Boostcamp AI Tech – Organizer & Mentor	2021 – 2022
Software Maestro (Ministry of Science and ICT, Korea) – Mentor	2022 – 2024
AI Weekly arXiv (Korea) – Moderator	2022 – 2025
PR12 Paper Reading Group	2019 – Present

EXPERIENCE

POSTECH	Pohang, Korea
<i>AI Research Scientist (Lab of Prof. Tae-Hyun Oh)</i>	2023 – 2024
Upstage (company behind the Solar LLM)	Seoul, Korea
<i>AI Contents & Education Manager / AI Pack User Guide Project Manager</i>	2022
LG Electronics (CTO Office AI Lab)	Seoul, Korea
<i>AI Research Engineer (Team led by Prof. Joseph Lim, USC)</i>	2020 – 2022
• Research topic: computer vision, scene graph generation • Collaborated with LG Toronto AI Lab on floor plan generation for object placement • Designed a driver action recognition system (Hyundai, Daimler)	
EASYWUM	Seoul, Korea
<i>Tech Lead, Smart IoT Indoor Garden Startup</i>	2016 – 2017

AWARDS & SCHOLARSHIPS

Korean Government-Funded Scholarship (Ph.D. Program), KAIST	2025 – Present
Best Student Presentation Award, IPIU (for the WACV 2026 paper [C6])	2025

INVITED TALKS

AI Talk for High School Sophomores and Juniors , Jeonbuk Science High School	2023
MODUCON: AI Community Organizing , Yonsei University / Lab for Everyone	2022
TensorFlow Everywhere Korea Keynote: What's New in TF 2.4 , Google	2021
Entrance Ceremony Speech , Seoul High School	2021
AI Talk for High School Sophomores and Juniors , Hanseong Science High School	2021
1st KICS Lecture on TF-based Deep Learning Programming , KICS	2018
Deep Learning: Algorithms, Software, and Systems , POSCO ICT	2017
AI Fundamentals , KT Education Center	2017

TEACHING EXPERIENCE

Deep Learning Project Management Process (Naver Boostcamp AI Tech) – Lecturer	2022
SW Health Contents Development (College of Nursing, Chung-Ang Univ.) – Lecturer	2018
Mobile Application Development – TA & Lecturer	2017, 2018
Programming Language – TA	2017

PATENTS

KR 10-1894314 (Granted): Method and Apparatus for Guiding a Recommended Parking Location (<i>tech. transferred, Nov. 2024</i>)
KR 10-2019-0005307: Method and System for Video Streaming Service
KR 10-2018-0089040: Recipient-oriented Transaction Verification Method and Apparatus for Preventing Double Spending
KR 10-2018-0089039: Proof-of-Probability (PoP) based Blockchain Compensation Method and Apparatus